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RESEARCH ARTICLE

Can zeolites be used in sugarcane cropping systems to help reduce nitrogen losses into the Australian Great Barrier Reef?

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Abstract

Aims The aim of this study was to determine the effectiveness of natural (NatZeo) and acid-treated (AcidZeo) zeolites in increasing the nitrogen (N) recovery of sugarcane grown under conditions highly conducive for N losses.

Methods This glasshouse trial replicated the pedo-climatic conditions typical of the Australian sugarcane industry in the Wet Tropics to evaluate the capacity and economic feasibility of using NatZeo and AcidZeo to reduce leaching and increase plant and soil N recovery in a highly permeable sandy-loam soil. Nitrogen fertiliser was applied at two rates (84 and 120 kg N ha^{-1}) as ^{15}N-enriched urea.

Results Compared with the unamended treatments, AcidZeo led to significantly higher average fertiliser N recoveries in both the soil (+24%) and plant (+54%), which resulted in significantly lower overall N losses (−22%). However, the significantly higher capacity of AcidZeo in retaining ions limited the availability of potassium and sulphur to the plants and led to a 4% yield reduction. The use of NatZeo led to lower abatements of N losses (−6%) but increased N recoveries in the plant (+22%). NatZeo did not result in as much cation immobilisation, which, combined with the increased plant/soil N recovery, resulted in the highest yield of the trial (+8% compared with the unamended treatments). The economic analysis indicated that only NatZeo could potentially increase farmer's income when applied in 20 cm wide strips around the fertiliser band.

Conclusions This study highlights that zeolites have potential to improve the environmental sustainability and profitability of sugarcane cropping systems.

Keywords Nitrogen use efficiency · Nitrate leaching · Water quality · Fertilisation · ^{15}N · Stable isotope

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Introduction

The Australian sugar industry needs rapid and innovative solutions to counter an increasing number of issues. Industry reports and scientific publications have highlighted clear trends of yield decline (Bell et al. 2015), soil acidification from high usage of nitrogen (N) fertiliser (Mthimkhulu et al. 2016) and profitability decreases due to low sugar prices and high production costs. At the same time, the public perception of the environmental impacts of sugarcane farming has rapidly deteriorated, and the sugar industry faces mounting pressure from government agencies and environmental groups to reduce the use of N fertilisers to mitigate the pollution of the Great Barrier Reef (Bell et al. 2015). There is therefore an urgent need to develop N management strategies that reduce N losses while maintaining productivity and profitability.

A growing body of literature has shown that increasing the capability of sugarcane cropping systems to retain fertiliser N in the soil-plant system is an essential step to improve water quality in tropical marine ecosystems, such as the Australian Great Barrier Reef (Thorburn et al. 2013, 2022; Deane et al. 2018). Importantly, an increase in N retention in the soil-plant system would offer the opportunity to decrease N fertiliser rates, and therefore improve the profitability of sugarcane cropping systems while reducing rates of soil acidification (Bell et al. 2015).

Nitrogen retention in the soil-plant system is comparatively low in sugarcane cropping systems, with plant N recovery rates usually not exceeding 40% of applied fertiliser N (Bell et al. 2015; Otto et al. 2016). Plant N recovery rates are particularly low in the Wet Tropics, where high levels of cloud cover substantially reduce sugarcane growth rates, lower the demand for N and therefore limit the plant's ability to promptly acquire fertiliser N from the soil. In these systems, the entire seasonal N rate must be applied before stalk growth starts and the crop reaches a height that prohibits trafficability (the 'out-of-hand' phase), which is typically at less than three months of age from emergence. At this age the crop contains only 10-20% of the maximum crop N uptake (achieved some 6 months later in the growing season), with much of the added fertiliser N then left in the soil and vulnerable to environmental loss. Nitrogen loss risks are exacerbated by the high-intensity rainfall events over the monsoonal wet season (December to April), which promote elevated N losses due to nitrate (NO_3^-) leaching and denitrification. Research on innovative N management strategies aimed at limiting N losses and increasing plant N recovery in sugarcane should therefore focus particularly on these regions with high potential N loss risk.

After been shown to increase water use efficiency (Gholizadeh-Sarabi and Sepaskhah 2013) and retain herbicides (Bakhtiary et al. 2013), the use of zeolites to limit N losses in agricultural soils has recently gained significant interest from the scientific community. Briefly, zeolites are mainly composed of negatively-charged clinoptilolite minerals and, in minor part of other positively-charged minerals (Khosravi et al. 2021). Thanks to their high cation exchange capacity (CEC – typically ranging between 50 and 100 cmol(+) kg^{-1}), zeolites found in nature (natural zeolites) retain readily exchangeable cations on the external mineral surfaces, and less readily exchangeable cations on a variety of sites within the zeolite structures. A proportion of these exchange sites, particularly in the internal mineral structure, are quite specific for cations with hydrated ionic radii of a particular size. For example, both potassium (K^+) and ammonium (NH_4^+) have similar, smaller sized hydrated ion radii that allow access to exchange sites that are quite specific for these sized ions. This specificity results in these cations becoming strongly adsorbed onto/into the zeolites and much less readily exchangeable with the soil solution. In the case of NH_4^+, this specific sorption can indirectly inhibit the nitrification of NH_4^+ to NO_3^- by reducing the availability of NH_4^+ for microbial transformations (Zwingmann et al. 2009; Omar et al. 2015). By reducing NO_3^- levels, natural zeolites reduce the risk of N loss via leaching or denitrification. By specifically adsorbing NH_4^+ cations, these products also limit the amount of soil mineral N that can be converted into the ammonia (NH_3) form at any time, potentially reducing volatilisation losses (Szerement et al. 2014).

However, the exchange sites on natural zeolites are not all selective for NH_4^+, meaning that the exchange sites can also adsorb other cations such as calcium (Ca^{2+}), magnesium (Mg^{2+}), sodium (Na^+) (as well as trace element cations such as zinc (Zn^{2+}) and copper (Cu^{2+})) from the soil solution. As a result, the actual capacity of natural zeolites to adsorb NH_4^+ cations is typically lower than 100

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cmol(+) kg^{-1} total exchange capacity, and will vary with the specificity of these exchange sites and the exposure to competing cations in the soil solution. To address these issues, a variety of chemical treatments of zeolites have been tested in an attempt to make them more selective for NH_4^+ adsorption, with acid treatment being particularly effective (Stocker et al. 2019). Acid-treated zeolites are obtained by washing natural zeolites with hydrochloric acid to remove Ca^{2+} and Mg^{2+} ions from the zeolite surface and replace them with hydrogen (H^+) ions. This operation is performed because the bonding strength between the negatively-charged sites of the zeolite and the NH_4^+ ions is lower than with Ca^{2+} and Mg^{2+} ions, but higher than with H^+ ions (lyotropic series). Acid treatment also removes lesser performing co-present minerals such as clay, dissolves parts of the zeolite mineral structure and exposes more of the internal exchange sites to more rapid exchange reactions. CEC often increases as a result of this, as the 'cage-like' structure of the zeolite becomes more ragged and/or ruptures, exposing more exchange sites to allow rapid cation sorption (and exchange). As a result, washing natural zeolites with acid tends to increase their CEC, and in the process, increases the number of readily accessible NH_4^+-selective sorption sites able to bind and retain NH_4^+ ions. Consequently, the CEC of acid-washed zeolites is approximately 200 cmol(+) kg^{-1} (when starting from a natural zeolite with a CEC of approximately 100 cmol(+) kg^{-1}) and require significantly lower application rates than natural zeolites.

To date, agricultural research has focused primarily on natural zeolites, which have been reported to effectively reduce N losses via NH_3 volatilisation (Ahmed et al. 2006), NO_3^- leaching (Piñón-Villarreal et al. 2013) and emissions of nitrous oxide (N_2O) and dinitrogen (N_2) (Zaman and Nguyen 2012), with consequent increases in crop N uptake (Ahmed et al. 2010; Ferretti et al. 2016). However, there is currently no literature that assess the use of different types of zeolites in high N-loss scenarios such as those in sugarcane cropping systems in the Wet Tropics, and even less on the use of acid-treated zeolites in agriculture. Importantly, the characteristic of zeolites of retaining N as NH_4^+ could prove to be particularly beneficial in sugarcane cropping systems, as it has been shown that sugarcane plants exhibit a preference for N uptake in the NH_4^+ form (Robinson et al. 2011).

Therefore, using Australian industry practices as an example, the aim of this study was to determine the effectiveness of natural and acid-treated zeolites on increasing the N recovery of sugarcane grown under conditions highly conducive for N losses. Conditions simulating the Australian Wet Tropics were reproduced in an eight-month glasshouse experiment to: i) determine whether natural and acid-treated zeolites have the potential to increase N recovery in the plant-soil system and reduce NO_3^- leaching in sugarcane cropping systems; and ii) inform future studies assessing the agronomic, environmental and economic implications of using natural and acid-treated zeolites at field scale. By establishing for the first time the potential of using zeolites to reduce the detrimental effects of low plant N recovery in sugarcane cropping systems, the results of this study will guide future research aimed at increasing the profitability and sustainability of the sugarcane industry.

Material and methods

Trial configuration

Natural and acid-treated zeolites were tested in a glasshouse trial using 5 m^3 of a sandy-loam Alfisol (Soil Survey Staff 2022), also known as Sodosol according to the Australian Soil Classification system (Isbell 2021). The reasons for this choice were two-fold: firstly, sandy-loam Alfisols occupy large portions of the Herbert region, one of the main cane growing districts in the Wet Tropics (ASRIS: Australian Soil Resource Information System n.d.); secondly, this soil type is highly conducive to NO_3^- leaching due its low CEC (varying between 1 and 5 cmol(+) kg^{-1}) and coarse texture. The soil (0-30 cm) was sourced from a commercial cane farm (25°02′39.6"S 152°22′40.5″E) that had the chemico-physical characteristics illustrated in Table 1. The soil was transported to a glasshouse facility, homogenised and then transferred into 28 large free-draining pots (0.06 m^3 capacity) to create a 40 cm-deep soil layer. The depth of the pot was deemed sufficient for the scope of this study since, in the first 8 months of plant growth (i.e. the length of this trial), over 70% of the root biomass of cane plants

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Table 1 Main soil physical and chemical properties (median ± standard error, $n = 4$) of the topsoil collected in Bundaberg, Queensland, Australia, and the Natural and Acid-treated zeolites

Material	Property	
Soil (0-30 cm)	pH (H_2O)	5.2 ± 0.05
	CEC (cmol(+) kg soil^{-1})	1.6 ± 0.1
	EC (μS cm^{-1})	21.1 ± 0.4
	Total C (%)	0.5 ± 0.05
	Total N (kg ha^{-1})	1018 ± 8
	Bulk density 0-30 cm (g cm^{-3})	1.38 ± 0.05
	Clay (%)	16
	Silt (%)	13
	Sand (%)	71
	Texture (USDA, Australian)	Sandy-loam
	Classification (USDA soil classification)	Alfisol
	Classification (Australian soil classification)	Sodosol
Natural zeolite	pH (H_2O)	5.5 ± 0.1
	CEC (cmol(+) kg soil^{-1})	105 ± 1.3
Acid-treated zeolite	pH (H_2O)	4.2 ± 0.1
	CEC (cmol(+) kg soil^{-1})	160 ± 1.8

is concentrated in the top 40 cm of soil (Smith et al. 2005).

The following treatments were imposed:

- NoZeo-0 N: soil not amended with zeolite and not amended with N fertiliser.
- NoZeo-84 N: soil not amended with zeolite and fertilised with equivalent of 84 kg N ha^{-1}. This N rate was 30% lower than the recommended N rate for application in the Herbert district (ca. 120 kg N ha^{-1}, dependent on soil organic matter content) calculated using the Six Easy Steps (6ES) approach (Sugar Research Australia 2013; Schroeder et al. 2019).
- NoZeo-120 N: soil not amended with zeolite and fertilised with 120 kg N ha^{-1}. The N rate was calculated using the 6ES approach.
- NatZeo-84 N: topsoil (0-10 cm) amended with natural zeolite and fertilised with equivalent of 84 kg N ha^{-1}.
- NatZeo-120 N: topsoil (0-10 cm) amended with natural zeolite and fertilised with equivalent of 120 kg N ha^{-1}.
- AcidZeo-84 N: topsoil (0-10 cm) amended with acid-treated zeolite and fertilised with equivalent of 84 kg N ha^{-1}.
- AcidZeo-120 N: topsoil (0-10 cm) amended with acid-treated zeolite and fertilised with equivalent of 120 kg N ha^{-1}.

Each treatment had four replications. Natural zeolite was directly sourced from Zeolite Australia Pty Ltd. (Millar et al. 2016), while the acid-treated zeolite was manufactured at the Queensland University of Technology, starting from natural zeolite from Zeolite Australia Pty Ltd. and following the procedure reported by Stocker et al. (2019). A detailed mineralogical analysis of the natural zeolite sourced from Zeolite Australia Pty Ltd. can be found in (Khosravi et al. 2021). Depending on the treatment, the soil for the top layer (0-10 cm) of each pot was amended with different quantities of natural (9.2 kg) or acid-treated (6.3 kg) zeolites to reach an overall CEC (soil + zeolite) of 50 cmol(+) kg soil^{-1}. The blending of soil and zeolite was conducted using a concrete mixer to achieve a homogenous mix. Using the methodology established by Campisi et al. (2016), the soil was placed into pots that were lined with a water-proof plastic bag equipped with a rubber drain pipe to collect the leachate (Fig. 1Sa). Each pot was planted with a two-node cane sett of the variety Q208(D), which is commonly used in the Herbert district (Sugar Research Australia 2018) (Fig. 1Sb). The pots were then housed in bigger non-draining pots and connected to an automated drip irrigation system (Fig. 1Sc). All pots were weeded manually every two weeks to avoid biases in the water and N budgets (Yang et al. 2013).

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Irrigation and fertilisation

The water application rates and air temperatures applied in the glasshouse were modified at monthly intervals and were designed to replicate the typical temporal patterns from August to March, the period of highest rainfall intensity in the Wet Tropics ("Am" climate according to the Köppen climate taxonomy). The temperatures and intensity, frequency and duration of the rainfall events were calculated using weather statistics for the last 10 years in Ingham, which were taken from the Australian Bureau of Meteorology (Fig. 1). Overall, the plants received the equivalent of 1841 mm over the duration of the trial.

To replicate local fertilisation practices, all pots (including the NoZeo-0 N treatment) were fertilised at planting with basal applications equivalent to 80 kg P ha^{-1} and 100 kg K ha^{-1} applied as mono potassium phosphate and muriate of potash. Using the methodology described in De Antoni Migliorati et al. (2014), N fertiliser was applied at rates equivalent to either 84 or 120 kg N ha^{-1} to fertilised treatments as a solution of 5% labelled ^{15}N urea and split between planting (30% of seasonal rate) and top dressing (70% of seasonal rate), which occurred three months after planting (84 days after planting - DAP).

Sampling and chemical analyses

Before starting the trial, the soil was analysed for texture (hydrometer method), CEC (1 M KCl extraction), electrical conductivity (EC; 1:5 extraction with de-ionised water), pH (1:5 extraction with de-ionised water), and total carbon (C; DUMAS combustion using LECO furnace at 1300 °C) and N content (DUMAS combustion using LECO furnace at 1300 °C). During the trial, the leachate accumulated under each pot was weighed and sampled on a fortnightly basis, starting after the first leaching event (85 DAP). The water samples were immediately refrigerated at 4 °C and analysed for NH_4^+-N and NO_3^--N content within two weeks using an AQ2+ discrete analyser (SEAL Analytical WI, USA).

The trial was terminated after eight months (241 DAP). Soil samples were collected from each pot at three depths (0-10 cm, 10-20 cm, 20-30 cm) and analysed for ^{15}N concentration (20–22 Isotope Ratio Mass Spectrometer, Sercon Limited, UK), NH_4^+-N and NO_3^--N content, and pH (1:5 extraction with de-ionised water). Plants and litter were collected from each plot and separated into roots, stems and leaves to determine above- and below-ground biomass production. Roots, stems and leaves were analysed separately to determine the concentrations of ^{15}N, macronutrients (total C, N, phosphorus (P), potassium (K), sulphur (S)) and micronutrients (total aluminium (Al), barium (B), Ca, copper (Cu), iron (Fe), Mg, manganese (Mn), Na and zinc (Zn)) in each plant compartment. It was not possible to analyse the soil for micronutrients as the samples used in previous analyses were destroyed during the Queensland floods that damaged the university's facilities in February 2022.

Calculations and statistical analysis

All calculations were conducted on an oven-dried basis. Nutrient uptake was calculated by multiplying the specific nutrient concentration in the above- and below-ground biomass by the dry matter weight of the above- and below-ground biomass, respectively. Total recovery

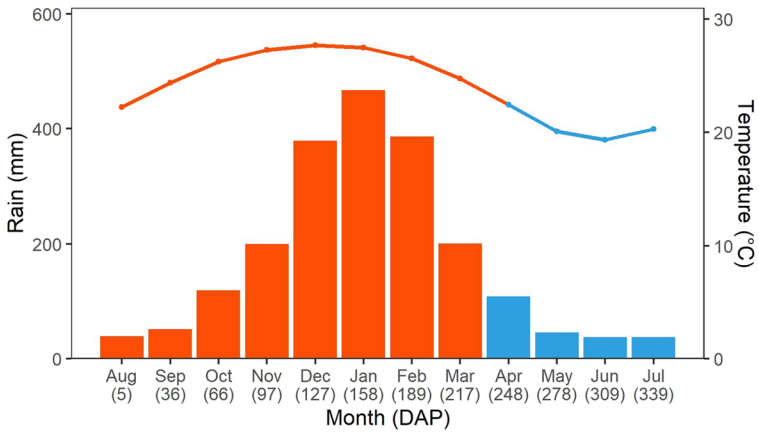

Fig. 1 Monthly rainfall (columns) and air temperature (line) patterns replicated in the experiment. Time is expressed as the equivalent month in the Wet Tropics (Herbert region) and days after planting (DAP). DAP coincides with the first day of each month. Monthly rainfall amounts and temperatures not replicated in the experiment are marked in blue

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of applied ^{15}N-labelled fertiliser was determined by mass balance.

The percentage of N derived from the labelled fertiliser (Ndff) in each plant and soil pool was determined using the following formula (IAEA 2001):

$$Ndff = \frac{\left(atom\%\ ^{15}N_{sample} - atom\%\ ^{15}N_{control}\right)}{\left(atom\%\ ^{15}N_{labelled\ fertiliser} - atom\%\ ^{15}N_{unlabelled\ fertiliser}\right)} \times 100 \quad (1)$$

where $atom\%\ ^{15}N_{\ sample}$ is the percentage of ^{15}N in plant and soil samples collected from each fertilised treatment, $atom\%\ ^{15}N_{\ control}$ is the percentage of ^{15}N in plant and soil samples collected from the unfertilised treatment (also known as natural abundance), $atom\%\ ^{15}N_{\ labelled\ fertiliser}$ is the percentage (5%) of ^{15}N in the labelled fertiliser (urea) and $atom\%\ ^{15}N_{\ unlabelled\ fertiliser}$ is the percentage of ^{15}N in the unlabelled fertiliser (urea).

Using Dobermann (2007) definition of partial nutrient balance, nitrogen use efficiency was expressed as the percentage of fertiliser ^{15}N recovered in the plant (Fixen et al. 2015). The percentage of ^{15}N recovered in the plant or soil pools was calculated using the following formula (IAEA 2001):

$$^{15}N\ recovery = \frac{atom\%\ ^{15}N\ excess_{recovered}}{atom\%\ ^{15}N\ excess_{applied}} \times 100 \quad (2)$$

where $atom\%\ ^{15}N\ excess_{\ recovered}$ is the percentage of ^{15}N (deducted of the atom% ^{15}N natural abundance) recovered in the plant or soil pool, and $atom\%\ ^{15}N\ excess_{\ applied}$ is the percentage of ^{15}N (deducted of the atom% ^{15}N natural abundance) applied with the ^{15}N enriched fertiliser.

Apparent N losses were calculated as the difference between applied N and the N recovered in leachate, soil and plant (above and below ground) fractions.

To enable the economic analysis and a more intuitive comparison between the stem weights measured in this glasshouse study (expressed as dry weight per pot) to those reported in field studies (expressed as fresh weight per-hectare), it was assumed a dry-to-wet weight conversion ratio of 4.5 and a standard planting density of 75,000 two-node setts ha^{-1} (Garside and Bell 2009). The limitations of this conversion are acknowledged and addressed in the Discussion section.

Differences in cane biomass production (inclusive of stems, leaves and roots), macro- and micronutrient concentrations, Ndff, N recovery and apparent N losses between treatments were assessed with the ANOVA and Fisher's least significant difference (LSD) post hoc tests using a confidence interval of 95%. Statistical analyses were performed within the R environment (R Development CoreTeam 2021).

Economic analysis

The *potential* economic implications of applying natural and acid-treated zeolite at field scale were estimated by taking into consideration the cost of natural and acid-treated zeolites, and the measured gains in stem weight generated by the application of zeolites. Economic returns were based only on a single cropping season following the application of zeolite (i.e. a plant crop). Using the abovementioned conversion, gains or penalties in commercial cane yields were estimated using the variations in stem weight measured in this study in response to the zeolite and fertiliser rates applied.

The extra income generated by the zeolite application was determined by the following formula:

$$Extra\ income_{za} = (Ay + Ay * Yg * CCS * Sp) - (Ay * Yg * CCS * Sp) \quad (3)$$

where Ay is the target cane yield of a plant crop grown in a sandy-loam soil in Australian Wet Tropical districts; Yg is the average yield gain (%) resulted from the application of natural or acid-treated zeolites compared to unamended soil; Sp is the average sugar price recorded in Australia in the last 3 years. Ay and Sp were estimated to be 120 t cane ha^{-1} (Calcino et al. 2019; De Antoni Migliorati et al. 2021) and 550 Australian dollars (AU$) t sugar^{-1} (QSL 2023), respectively. For the purpose of calculating the income for the farmer, CCS (Commercial Cane Sugar) was set to 9%, since typically CCS is approximately 13% (Calcino et al. 2019; De Antoni Migliorati et al. 2021) but 4% is owned by the mill.

The cost of zeolite application ($Cost_{za}$) at field level was calculated with the following formula:

$$Cost_{za} = Zp * Ar * Fa \quad (4)$$

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where *Zp* is the estimated commercial price of zeolite, inclusive of transport; *Ar* is the zeolite application rate; and *Fa* the percentage of the field area amended with zeolite. *Zp* was estimated to be AU$ 240 t^{-1} and AU$ 340 t^{-1} for natural and acid-treated zeolite, respectively (Zeolite Australia Pty Ltd., personal communication).

In this study, the zeolite application rates (*Ar*) designed to achieve an overall CEC (soil + zeolite) of 50 cmol(+) kg $soil^{-1}$ in the topsoil (0-10 cm) over 100% of the field area ($Fa = 100\%$) were 319 t ha^{-1} and 166 t ha^{-1} for natural and acid-treated zeolite, respectively. In the event that only the bed was amended with zeolite ($Fa = 50\%$), *Ar* for natural and acid-treated zeolite was determined to be 159 t ha^{-1} and 83 t ha^{-1}, respectively. In the case of only the area around the fertiliser band being amended with zeolite ($Fa = 10\%$, band width = 20 cm), *Ar* for natural and acid-treated zeolite was determined to be 32 t ha^{-1} and 17 t ha^{-1}, respectively. $Cost_{za}$ did not include application costs as preliminary field trials indicated that zeolites could be applied with conventional fertiliser spreaders at a negligible cost for the farmer.

The number of cane seasons required to recoup the costs of zeolite application at field level (Cane $seasons_{za}$) was assessed with the following formula:

$$Cane\ seasons_{za} = \frac{Extra\ income_{za}}{Cost_{za}} \quad (5)$$

This formula assumed that the zeolites continue to provide the same N retention benefits in each and every year after application, which is an assumption that will need to be tested under field conditions.

Results

Crop growth

Emergence was observed 29 DAP, with no differences among treatments. At the end of the 8-month trial, the total biomass production of the cane plants showed a significant difference only between the unfertilised treatment and the fertilised ones (Fig. 2). On average, total dry matter (DM) in the treatments fertilised with 84 kg N ha^{-1} and 120 kg N ha^{-1} was 46% and 51% higher than in the unfertilised control, respectively (Table 2). Across replicates, the average total biomass production (inclusive of stems, leaves and roots) in the control was 317 g DM $plant^{-1}$ and varied from 430 g DM $plant^{-1}$ (AcidZeo-84 N) to 511 g DM $plant^{-1}$ (NatZeo-120 N) in the fertilised treatments. Across N fertiliser rates, total biomass in the NatZeo and AcidZeo treatments was 59% and 39% higher than in the NoZeo-0 N, respectively. On a per-hectare basis, the stem fresh weights measured in this study would have been equivalent to between 71.8 tc ha^{-1} (NoZeo-0 N) and 104.7 tc ha^{-1} (NatZeo-84 N) (Table 2). The unit "tc", commonly

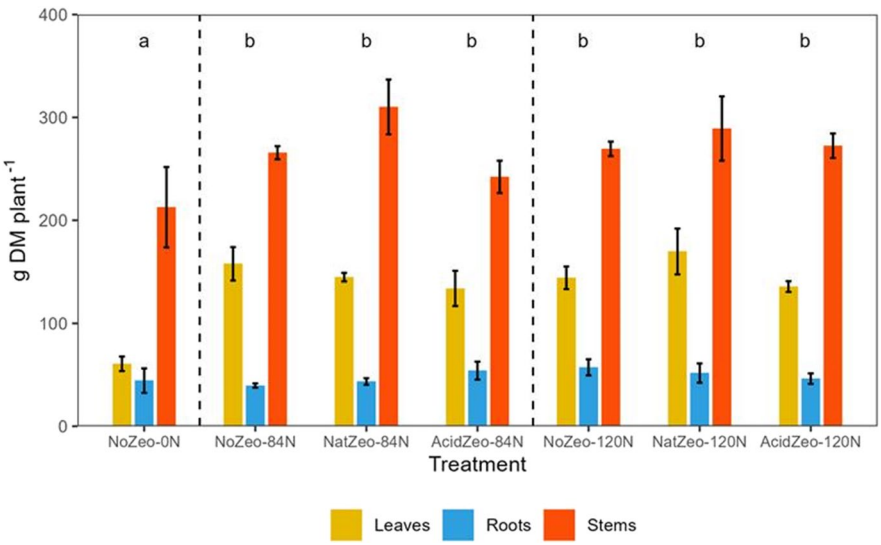

Fig. 2 Leaf, root and stem weights, expressed as dry matter (DM), measured at the end of the 8-month experiment. Bars indicate standard errors. Means denoted by a different letter indicate significant differences in total biomass ($p < 0.05$)

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Table 2 Recovery of fertiliser ^{15}N measured at the end of the investigation (mean ± standard error, $n = 4$)

Treatment	Total biomass [g DM plant^{-1}]	Stem weight [tc ha^{-1}]	^{15}N recovered in Plant [%]	^{15}N recovered in Soil [%]	^{15}N lost via leaching [%]	Unaccounted ^{15}N [%]	Overall ^{15}N recovered in Plant-Soil system [%]	Overall ^{15}N lost [%]	Ndff [%]
NoZeo-0 N	317.7 ± 52^{a}	71.8 ± 10^{a}	–	–	–	–	–	–	–
NoZeo-84 N	462.9 ± 13^{b}	89.7 ± 2.7^{abc}	17.5 ± 1.3^{ab}	21.0 ± 0.6^{a}	30.7 ± 6.1^{a}	30.8 ± 5.9^{a}	38.5 ± 1.2^{a}	61.5 ± 1.2^{c}	19.5 ± 4.1^{a}
NatZeo-84 N	498.3 ± 25^{b}	104.7 ± 6.9^{c}	20.8 ± 3.3^{abc}	20.8 ± 1.9^{a}	31.7 ± 4.1^{a}	26.7 ± 10.7^{a}	41.6 ± 5.1^{ab}	58.4 ± 5.1^{bc}	23.8 ± 2.8^{b}
AcidZeo-84 N	430.1 ± 34^{b}	81.7 ± 7^{ab}	23.9 ± 2.9^{bc}	26.2 ± 5.3^{a}	25.7 ± 6.2^{a}	24.2 ± 3.7^{a}	50.1 ± 4.9^{bc}	49.9 ± 4.9^{ab}	27.1 ± 1.6^{bc}
NoZeo-120 N	470.8 ± 17^{b}	90.9 ± 3.1^{abc}	16.1 ± 0.8^{a}	20.7 ± 2.9^{a}	29.6 ± 10.3^{a}	33.5 ± 4.7^{a}	36.9 ± 2.1^{a}	63.1 ± 2.1^{c}	23.6 ± 2.5^{b}
NatZeo-120 N	510.6 ± 54^{b}	97.6 ± 13^{bc}	20.2 ± 3.5^{abc}	21.5 ± 3.2^{a}	32.6 ± 3.4^{a}	25.7 ± 4.6^{a}	41.7 ± 3.1^{ab}	58.3 ± 3.1^{bc}	30.0 ± 2.4^{c}
AcidZeo-120 N	454.4 ± 18^{b}	91.9 ± 3.4^{abc}	27.5 ± 2.4^{c}	25.6 ± 2.7^{a}	27.0 ± 3.9^{a}	19.9 ± 5.3^{a}	53.1 ± 4.6^{c}	46.9 ± 4.6^{a}	37.1 ± 2.0^{d}

Total biomass includes stems, leaves and roots. To enable a comparison with commercial yields, stem weight was converted and expressed as fresh weight in tons of fresh cane (tc) per hectares. N lost via leaching includes both NH_4^+ and NO_3^-. Overall N losses are the sum of unaccounted N and N lost via leaching. Different letters denote statistically significant differences among treatments ($p = 0.05$)

used in the sugarcane industry, stands for "tonnes of cane" and indicates the fresh weight of sugarcane at harvest.

On average, the use of natural zeolites (NatZeo treatments) increased stem weight by 12% compared with the treatments fertilised with the same N rate but not amended with zeolite (NoZeo treatments) (Fig. 2). In contrast, AcidZeo treatments recorded an average 4% and 10% penalty in stem weight compared with the NoZeo and NatZeo ones, respectively. Within the same zeolite treatment, stem weight and total biomass production tended to be higher in the 120 N than in the 84 N treatments, although never significantly (Table 2).

Nitrogen losses via leaching

The collection of leachate samples started 85 DAP, when the first leaching event occurred. The lack of leaching during the first three months of the trial was due to the low irrigation regimes applied, aimed at mimicking the dry winter months between August and November in the Herbert region (Fig. 1).

Cumulative amounts of NH_4^+ found in the leachate samples did not change substantially across treatments or N fertilisation rates, either in terms of temporal dynamics or in terms of absolute numbers (Fig. 3). At the end of the eight months of the experiment, cumulative NH_4^+ losses across treatments were on average 9.2 kg N- NH_4^+ ha^{-1} and varied from 7.6 kg N- NH_4^+ ha^{-1} (AcidZeo-84 N) to 11.0 kg N-NH_4^+ ha^{-1} (AcidZeo-120 N). However, cumulative amounts of NO_3^--N lost via leaching instead varied significantly over time and across treatments (Fig. 3). In the fertilised treatments, high NO_3^- losses were recorded in the first eight weeks (85-143 DAP) after top dressing (84 DAP), but then plateaued to rates equivalent to the unfertilised Control treatments. On average, treatments fertilised with 84 kg N ha^{-1} and 120 kg N ha^{-1} lost a total of 15.9 kg N-NO_3^- ha^{-1} and 25.8 kg N-NO_3^- ha^{-1} via NO_3^- leaching, respectively (Fig. 3).

At both 84 N and 120 N fertiliser rates, the treatments amended with acid-treated zeolite had the lowest NO_3^- losses, statistically not different from the unfertilised treatment ($p > 0.05$) (Fig. 3). When the soil was amended with natural zeolite, the cumulative amounts of NO_3^- lost were also not dissimilar from those measured in the unamended treatments. At equivalent N rates, the use of acid-treated zeolite reduced total N losses via leaching (sum of NH_4^+ and NO_3^-) by 16% (84 N) and 9% (120 N) compared with the NoZeo treatment (Table 2).

Nitrogen budget

On average, only 44% of the applied fertiliser N was recovered in the plant-soil system, while the remaining 56% was lost via either leaching (30%) or could not be accounted for (26%) (Table 2). The fertiliser N recovered in the plant was on average 21% of the total amount applied, varying between 16% (NoZeo-120 N) and 28% (AcidZeo-120 N) (Fig. 4). Across treatments, the average fertiliser N recovery in the soil

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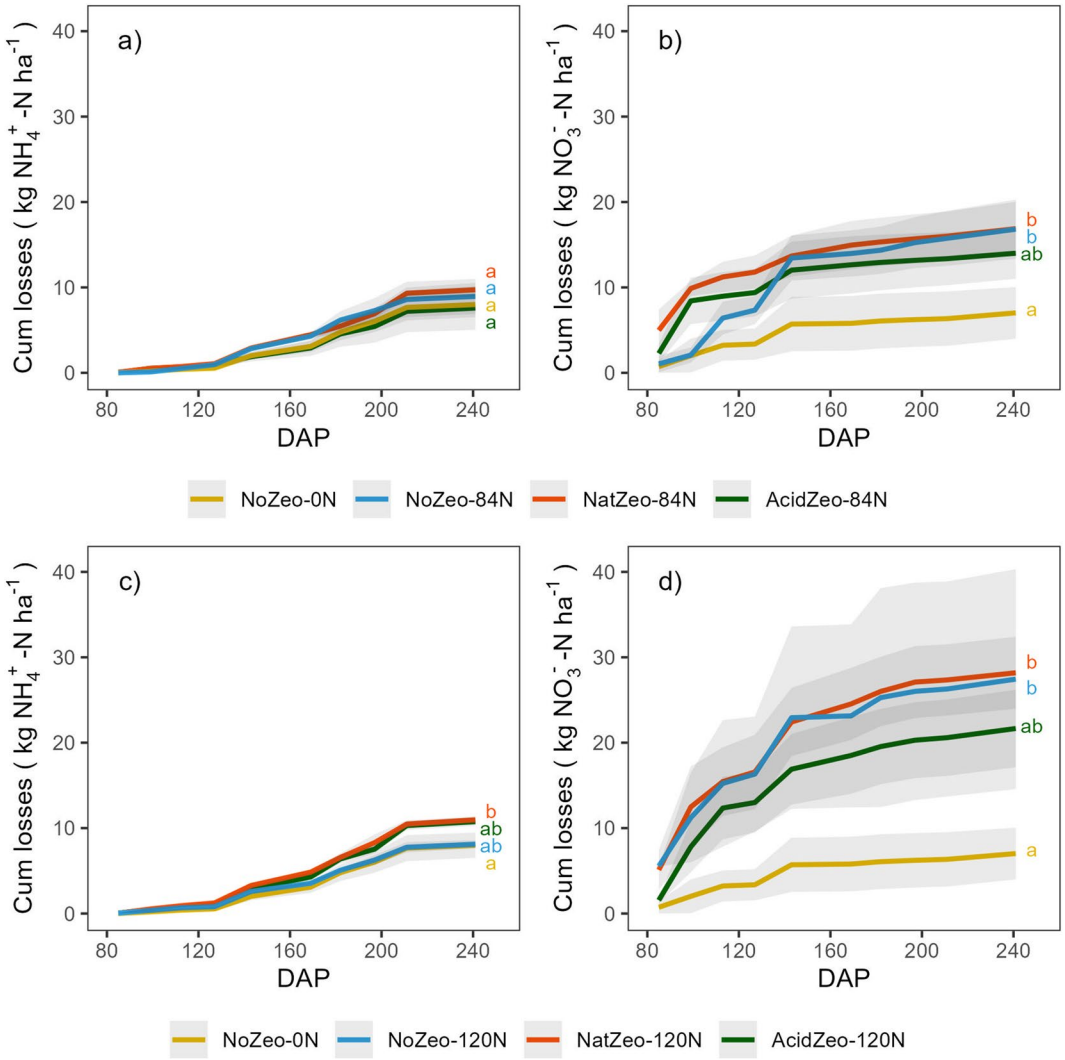

Fig. 3 Cumulative NH_4^+ (left) and NO_3^- (right) losses via leaching measured in the water drained out of the pots with or without zeolites and fertilised with 84 kg N ha^{-1} (**a**, **b**), or with 120 kg N ha^{-1} (**c**, **d**). In each graph the mineral N levels measured in the 84 N and 120 N rates treatments are compared with the control (No-Zeo-0 N). Shaded areas indicate standard errors, while lines denoted by a different letter indicate significant differences in cumulative N losses at the end of the experiment ($p < 0.05$)

was 23% and varied between 21% (NoZeo-120 N) and 26% (AcidZeo-84 N) (Fig. 4). Total N losses via leaching ranged between 26% (AcidZeo-84 N) and 33% (NatZeo-120 N) of applied N, with an average of 30% across treatments. The amount of unaccounted fertiliser N (representing cumulative N loss via other pathways) was on average 27% and varied from 20% (AcidZeo-120 N) and 34% (NoZeo-120 N).

Overall, there was a consistent tendency of acid-treated zeolite to substantially improve N retention in the plant-soil system compared with natural zeolite and the unamended soil (Table 2). In the plants grown in soil amended with acid-treated zeolites, Ndff values were significantly higher compared with those grown in unamended soil and tended to be higher than those grown in soil amended with natural zeolite (Table 2, Fig. 4). At both N fertiliser rates, acid-treated zeolites significantly increased the amount of N recovery in the plant-soil systems compared to the unamended treatments ($p < 0.05$, Table 2). This dynamic was

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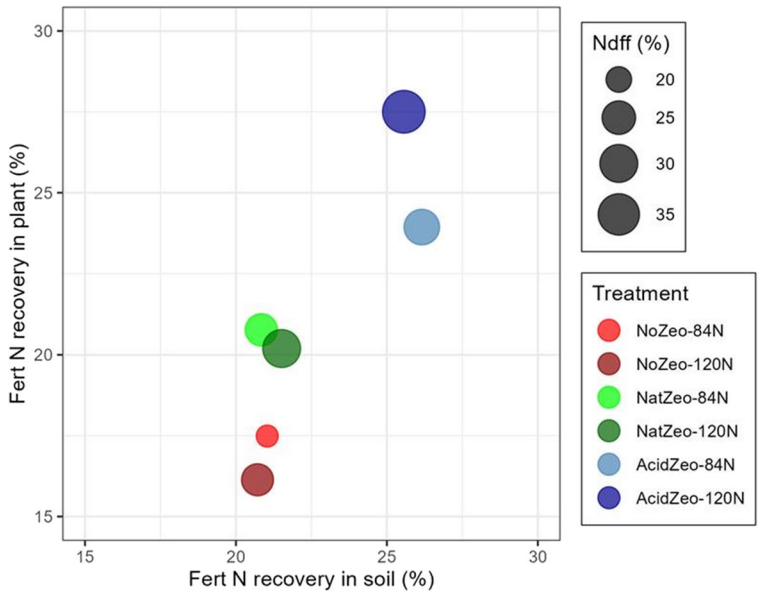

Fig. 4 Effects of zeolite type (natural or acid-treated) on the percentage of fertiliser N recovered in the plant and soil pools, as well as on the percentage of plant N derived from fertiliser (Ndff) across different N fertiliser rates (84 kg N ha^{-1} or 120 kg N ha^{-1})

reflected in the overall N losses (inclusive of N losses in leachate and the total fertiliser N not accounted for in the soil and plant fractions), which in the AcidZeo treatments were significantly lower than in the NoZeo equivalents ($p < 0.05$, Table 2).

During this eight-month study, the application of natural zeolite reduced the average amount of fertiliser N lost from the soil-plant system by 6% compared with the unamended soil, while the acid-treated zeolite reduced losses by 22% (Table 2). The analysis of apparent N losses indicated that, regardless of the fertiliser N rate applied, on average only 52% of apparent total fertiliser N loss could be attributed to NO_3^--N leaching (Table 2).

The CEC analysis of the soil layers sampled at harvest showed that, across all treatments, the CEC of the soil + zeolite mixes in the top 10 cm layer had remained close to 50 cmol(+) kg soil^{-1}. The CEC of the lower layers had remained similar to that of the unamended soil (on average 1.17 cmol(+) kg soil^{-1} and 0.88 cmol(+) kg soil^{-1} for the natural and acid-treated zeolites, respectively; data not shown). Consistent with this result, the majority of the fertiliser N retained in the soil was recovered in the top 10 cm, with treatments amended with acid-treated zeolite tending to have higher fertiliser N recoveries in this 10 cm layer than treatments with natural zeolite or without zeolite (Fig. 5).

Macro- and micronutrient analysis

The analysis of macro- and micronutrient concentrations in the stems showed reductions in tissue concentrations across most nutrients, which was consistent with the effect of plant growth on nutrient dilution. However, there was evidence that acid-treated significantly limited plant access to K and, to a smaller degree, S compared to the unamended treatment. A similar trend was observed for natural zeolites, even though differences were less substantial. Compared to treatments unamended with zeolites, the plant uptake of K averaged across N fertiliser rates was reduced by 35% (acid-treated zeolites) and 11% (natural zeolites), while the plant uptake of S was reduced by 16% (acid-treated zeolites) and 2% (natural zeolites) (Table 3). Tissue concentrations of P and micronutrients (Al, B, Ca, Cu, Fe, Mg, Mn, Na and Zn) did not differ statistically across treatments.

Economic analysis

If the 17% and 7% stem weight increase due to application of natural zeolite in the 84 N and 120 N treatments (Table 2) translated to equivalent yield increases in the field (*Yg*), these treatments would have resulted in an estimated *Extra income*$_{za}$ of approximately + AU$ 1000 ha^{-1} and + AU$ 400 ha^{-1}

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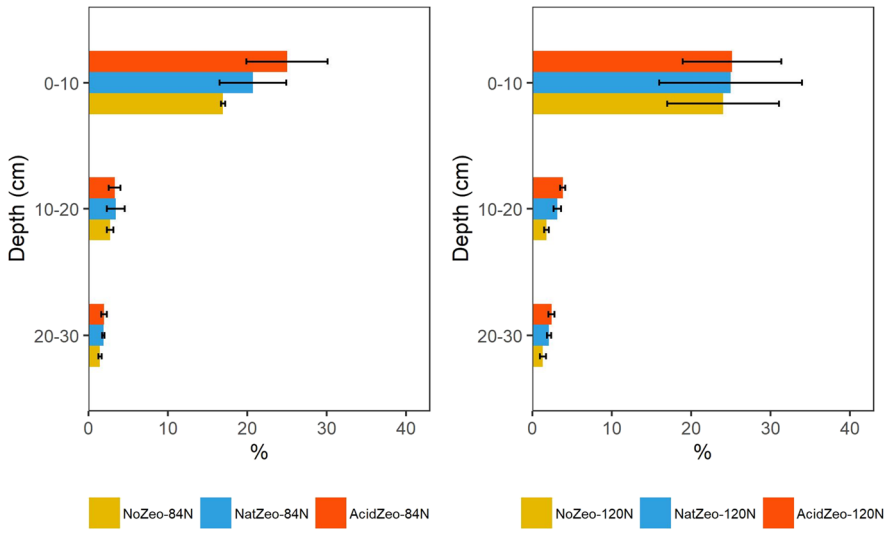

Fig. 5 Percentage of ^{15}N-labelled fertiliser recovered in the soil by depth increment across the different treatments fertilised with 84 kg N ha^{-1} (left) and 120 kg N ha^{-1} (right). Error bars indicate standard errors

Table 3 Left: total uptake of K, P, S in the plant tissues (sum of nutrient uptake in the above- and below-ground biomass) at the end of the experiment (mean ± standard error, $n = 4$) Right: total plant uptake of K, P, S averaged across N fertiliser rates. Different letters denote statistically significant differences

Treatment	Nutrient uptake per plant			Treatment	Nutrient uptake averaged across N fertiliser rates		
	K [g plant^{-1}]	P [g plant^{-1}]	S [g plant^{-1}]		K [g plant^{-1}]	P [g plant^{-1}]	S [g plant^{-1}]
NoZeo-0 N	1.32 ± 0.09^{bc}	0.31 ± 0.004^{a}	0.28 ± 0.01^{a}	NoZeo	1.34 ± 0.09^{b}	0.31 ± 0.01^{a}	0.31 ± 0.02^{b}
NoZeo-84 N	1.21 ± 0.05^{abc}	0.28 ± 0.01^{a}	0.31 ± 0.01^{ab}	NatZeo	1.21 ± 0.11^{b}	0.34 ± 0.02^{a}	0.31 ± 0.01^{b}
NatZeo-84 N	1.23 ± 0.04^{abc}	0.34 ± 0.03^{a}	0.3 ± 0.01^{ab}	AcidZeo	0.9 ± 0.06^{a}	0.3 ± 0.01^{a}	0.26 ± 0.01^{a}
AcidZeo-84 N	0.82 ± 0.07^{a}	0.3 ± 0.01^{a}	0.26 ± 0.01^{a}				
NoZeo-120 N	1.47 ± 0.24^{c}	0.32 ± 0.02^{a}	0.35 ± 0.04^{b}				
NatZeo-120 N	1.19 ± 0.24^{abc}	0.34 ± 0.03^{a}	0.31 ± 0.02^{ab}				
AcidZeo-120 N	0.98 ± 0.09^{ab}	0.31 ± 0.02^{a}	0.27 ± 0.01^{a}				

compared to cane grown on unamended soil, respectively.

On the other hand, the −9% and + 1% variations in stem weight measured after the application of acid-treated zeolite in the 84 N and 120 N treatments (Table 2) would have led to an estimated change in *Extra income*$_{za}$ equivalent to - AU$ 500 ha^{-1} (income loss) and + AU$ 60 ha^{-1} (income gain), respectively.

In the case of natural zeolite, the cost of zeolite application at field level (*Costs*$_{za}$) would have varied between AU$ 77,000 ha^{-1}, AU$ 38,000 ha^{-1} and AU$ 7600 ha^{-1} if the zeolite had been applied on the entire field area ($Fa = 100\%$), just on the bed ($Fa = 50\%$) or on an approximately 20 cm wide strip around the fertiliser band ($Fa = 10\%$), respectively (Table 4). In the case of acid-treated zeolite, such costs would have varied from AU$ 56,000 ha^{-1}, AU$ 28,000 ha^{-1} and AU$ 5000 ha^{-1}, respectively.

Given the yield gains observed with natural zeolite in the 84 N treatments, the number of cane seasons required to recoup the costs of natural zeolite application (*Cane Season*$_{za}$) varied from almost 80 ($Fa = 100\%$) to 8 ($Fa = 10\%$) (Table 4). In the case of natural zeolite in the 120 N treatments, *Cane Season*$_{za}$ varied from almost 180 ($Fa = 100\%$) to 18 ($Fa = 10\%$). The negligible yield gains measured with acid-treated zeolite in the 120 N treatments resulted in *Cane Season*$_{za}$ varying from approximately 800 ($Fa = 100\%$) to

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Table 4 *Potential* economic implications of applying natural and acid-treated zeolite at field level to increase the CEC of the topsoil (0-10 cm) from 1.6 cmol(+) kg soil^{-1} to 50 cmol(+) kg soil^{-1}

	Natural zeolite			Acid-treated zeolite		
Fa (%)[a]	100	50	10	100	50	10
$Cost_{za}$ (AU$)[b]	76,471	38,235	7647	56,460	28,230	5646
Cane $Season_{za}$ 84N[c]	77	38	8	–	–	–
Cane $Season_{za}$ 120N[c]	176	88	18	854	427	85

[a]Fa: percentage of the field area amended with zeolite

[b]$Cost_{za}$: cost of zeolite adoption at field level

[c]Cane $Season_{za}$: number of cane seasons required to recoup the costs of zeolite application

80 (*Fa* = 10%) (Table 4). *Cane* $Season_{za}$ was not calculated for acid-treated zeolite in the 84 N treatment since it led to a yield penalty compared to the unamended soil.

Discussion

The results of this study identified that natural zeolite was effective at increasing plant N recovery, stem weight and total biomass, but less effective at substantially reducing N losses. On the other hand, acid-treated zeolite demonstrated a near-significant ability to increase plant N recovery and reduce N losses, but had a slightly detrimental effect on stem weight and biomass production. Since the present study aimed to assess the potential environmental, agronomic and economic benefits of using zeolites, as well as to identify key research areas to guide future investigations, this section is organised around four key research questions: i) could the results obtained in this study be considered representative of field conditions? ii) are acid-treated zeolites too effective at retaining N in the soil? iii) are natural zeolites the best compromise? iv) can the use of zeolites be economically viable in sugarcane cropping systems? This section ends with considerations on the limitations of this study.

The glasshouse trial replicated well field conditions

Considering the highly permeable soil used in this experiment, the plant and soil N dynamics observed in the unamended treatments were in good agreement with those measured in field trials conducted under similar conditions. The average stem weight measured after eight months in the NoZeo-84 N and NoZeo-120 N treatments would have been equivalent to yields of 89.7 tc ha^{-1} and 90.9 tc ha^{-1}, respectively (Table 2). These values were equivalent to 75% and 76% of the yield potential (120 tc ha^{-1}) of a mature crop (approximately 11 months) grown under ideal soil conditions in the Australian Wet Tropics (Calcino et al. 2019). Importantly, the yields measured in the unamended treatments of this glasshouse trial were in line with those measured by Salter et al. (2015) during a field experiment conducted under very similar pedo-climatic conditions. In their study, carried out in the Wet Tropics on a sandy loam soil, Salter et al. (2015) reported cane yields averaging 92 tc ha^{-1} with a fertiliser N rate of 140 kg N ha^{-1}.

The average N recoveries in the plant (17%) and soil (21%) matrices measured in the unamended treatments of this study (Table 2) were comparable to those measured by Takeda et al. (2021), the only field experiment conducted under very similar pedo-climatic conditions using ^{15}N stable isotope techniques. Takeda's 10-month field experiment was conducted on a commercial sugarcane farm on a sandy clay loam soil in the Australian Wet Tropics, and the sugarcane plants were fertilised with 150 kg N ha^{-1} as urea. During that experiment, Takeda et al. (2021) observed that the amount of fertiliser N recovered in the plant and soil matrices amounted to 16% and 37%, respectively. The N losses reported by Takeda et al. (2021) tended to be lower (47%) than those measured in this study in the NoZeo-84 N (62%) and NoZeo-120 N (63%) treatments (Table 2). These differences could be explained by the slightly drier weather conditions (1418 mm during the experimental period) and finer texture of the soil where Takeda et al. (2021) conducted the experiment, which are likely to have led to lower N leaching losses and slightly

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higher cane yields (105 tc ha^{-1}). Also, given that the pots were left draining for 15 days between leaching sample collection events, it is probable that between sampling events a percentage of N in the solution was converted to N_2 and lost in the atmosphere, and therefore was not accounted for in this study.

At a general level, the N losses measured in the unamended treatments of this study were in good agreement with those reported across the sugarcane industry. In Brazil, Boschiero et al. (2018) and de Castro Gava et al. (2005) reported fertiliser N losses of 40–50%, while in Australia, Chapman et al. (1994), Vallis et al. (1996), Prasertsak et al. (2002) and Takeda et al. (2022) observed fertiliser N losses varying between 50% and 70%. The pedo-climatic conditions replicated in this glasshouse trial can therefore be considered representative of field settings, and the treatment results extrapolated with reasonable confidence.

Are acid-treated zeolites too effective at retaining N in the soil?

The greater CEC of the acid-treated zeolite (160 cmol(+) kg $soil^{-1}$, Table 1) was very effective in retaining the fertiliser N in the rhizosphere under environmental conditions that were highly conducive for significant N losses via leaching (Fig. 3), which are typical of Wet Tropical summers. The use of acid-treated zeolite led indeed to significantly higher average fertiliser N recoveries in both the soil (average across fertiliser rates: +24%, Table 2, Fig. 4) and plant (average across fertiliser rates: +54%, Table 2, Fig. 4) matrices compared with the unamended fertilised treatments, which resulted in significantly lower overall N losses (average across fertiliser rates: −22%) (Table 2). Accordingly, the amount of N derived from the fertiliser (Ndff) in the plants grown in soil amended with acid-treated zeolites was always significantly higher than those measured in the unamended treatments (Fig. 4). However, the increased fertiliser N retention in the plant-soil system did not translate into increased stem weight, which in the AcidZeo treatments was on average 4% and 10% lower than in NoZeo and NatZeo, respectively (Table 2, Fig. 2).

The reduction in total biomass and stem weight observed in the AcidZeo treatments was likely to have been due to acid-treated zeolites being too effective in retaining K and, to a lesser degree S. Plant tissues from the treatments amended with acid-treated zeolites had significantly less K and S in the stems than in the NoZeo treatments (Table 3). Even though there was a tendency for dilution of tissue nutrient concentration with increasing biomass for K and S (which was to be expected given the finite amount of available fertiliser added), the analysis of nutrient uptake averaged across N fertiliser rates showed that K and S concentrations in the acid-treated zeolite treatments were significantly lower than expected on the basis of plant growth/dilution. This nutrient deficiency could have been due to the acidification of the top soil layer caused by the rather low pH of the acid-treated zeolite (4.2, see Table 1), as numerous studies have reported a reduction in K and S availability to plant uptake in soils with a pH lower than approximately 5 (El-Tahir 1976; Roques et al. 2013).

However, the main reason of this nutrient deficiency is hypothesised to be due to the higher CEC of the acid-treated zeolite and the proportion of those negatively-charged clinoptilolite mineral exchange sites that strongly sorbed K cations. Interestingly, other positively charged non-clinoptilolite minerals also present in the zeolite granules appeared to be able to sorb S anions, which ultimately would have limited the availability of these nutrients to the plants.

As a result, plants in the AcidZeo treatments did not reach potential yield due to limited availability of K (with possible secondary limitations of S) rather than because of reduced N availability, with high N concentrations in the plant tissues indicating that N was never a limiting factor. It was not possible to corroborate this hypothesis by assessing K, P and S levels in the soil-zeolite mix since these samples were lost during the 2022 Queensland floods.

This ‘excessive’ efficiency of surface-modified zeolites in retaining nutrients in agroecosystems has not previously been reported in the literature, nor even in the recent extensive reviews produced by Jarosz et al. (2022) and Mondal et al. (2021). The only exception seems to be Lee et al. (2019), who reported reduced rice yields due to N and P immobilisation in soils amended with zeolites. The lack of reports of negative effects of natural or surface-modified zeolites on nutrient availability and plant development might be due to the ‘publication bias’, that is a well-known tendency in peer reviewed research studies, to mainly report successful experiments to facilitate the publication process (Mlinarić et al.

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2017). Future research on nutrient immobilisation in soil amended with zeolites is therefore warranted.

Are natural zeolites the best compromise?

Even though less remarkable than those observed in the AcidZeo treatments, the use of natural zeolites led to substantial abatement of overall N losses (average across fertiliser rates: −6%, Table 2) and increases in plant N uptake (average across fertiliser rates: +22%, Table 2) compared with the unamended fertilised treatments (Fig. 4). Importantly, the comparatively lower CEC of natural zeolites (105 cmol(+) kg soil^{-1}, Table 1) and neutral pH (Table 1) did not lead to excessive nutrient immobilisation (Table 3), which, combined with the increased N retention in the plant-soil system, allowed production of the highest stem weights in the trial (average across fertiliser rates: +12% compared with the unamended treatments) (Fig. 2). As a result, fertiliser N recoveries in the natural zeolite treatments were higher than those recorded by Takeda et al. (2022) in the Dry and Wet Tropics, where plants were grown under more favourable pedo-climatic conditions but the soil was not amended with zeolites.

The agronomic and environmental performance of zeolites could be highly relevant for the sugarcane industry in the Wet Tropics, where significant amounts of fertiliser N accumulate in the soil and become susceptible to nitrification and denitrification in periods characterised by high-intensity rainfall events (January to April). As a result, in the Wet Tropics elevated amounts of fertiliser are lost every season via NO_3^- leaching, runoff and emissions of N_2 and N_2O (Bell et al. 2015).

The reduction of current fertiliser N rates is of high importance for the sustainability and therefore the future of the sugarcane industry. Even at recommended N rates, 50% of applied N can be lost to the environment due to the mismatch between fertiliser N availability and plant N demand (Takeda et al. 2022), and it has been proposed that current N fertiliser rates could therefore be reduced by 25-30% without incurring any yield penalty (Thorburn et al. 2011; Takeda et al. 2021). However, the feasibility of such strategies will depend entirely on the ability to reduce N losses whilst simultaneously increasing the ability of the crop to acquire fertiliser N. Natural zeolites would appear to be a strategy that may potentially achieve these outcomes. The increased N retention in the plant-soil system attainable with natural zeolites could therefore be used to safely reduce current N fertiliser rates, especially when considering that soil N mineralisation plays a major role in these systems. As established by Takeda et al. (2021) and confirmed in Takeda et al. (2022) with a mix of ^{15}N tracer and greenhouse gas measuring chamber techniques, 65-75% of total plant N uptake in tropical sugarcane systems is derived from the mineralisation of native soil N. Thanks to the ability of natural zeolites to retain N, fertiliser N rates could therefore be reduced to allow soil mineralisation to sustain the majority of crop N demand during the early phases of the plant growth. It is during these early phases (2-4 months after planting) that the majority of fertiliser N losses occur, when the crop is not sufficiently developed to effectively acquire the fertiliser N (Zhao et al. 2017). With effective use of natural zeolites, fertiliser N would be retained close to the rhizosphere and released gradually, allowing fertiliser N to be used mainly to replenish the soil N mineralised during the season. This strategy would avoid the degradation of soil fertility and reduce N losses due to asynchronicity between crop N demand and fertiliser supply.

The feasibility of using natural zeolites to safely reduce N fertiliser rates in the short term was confirmed by this study, where the stem weights in the treatment fertilised with 30% less N than the district average (84 kg N ha^{-1}) were equivalent to those in the treatment fertilised according to the district average (120 kg N ha^{-1}). Given the potential implications for the sustainability and profitability of the sugarcane industry, future research should investigate the long-term feasibility of using natural zeolites in reducing current fertiliser N rates by means of multi-season field trials across different pedoclimatic regions.

Can the use of zeolites be economically viable in sugarcane cropping systems?

This study showed that the use of zeolites in sugarcane systems can potentially lead to substantial increases in the farmer's income, with these increases being enough to recoup the costs associated with zeolite application in a few growing seasons. In practical terms, this was only true for natural zeolites when applied in 20 cm wide strips around the fertiliser band (Fa = 10%) since, on

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average, the time to recoup the cost of this practice was estimated to vary between 8 and 18 growing seasons (Table 4).

It has to be taken into consideration that in commercial settings the application of zeolites is hypothesised only to occur at the planting of the plant crop and that, in Australia, a sugarcane cropping cycle typically lasts 4-5 seasons (De Antoni Migliorati et al. 2021). Therefore, the time to recoup the costs of zeolite applications (payback period) could vary between 2 and 4 crop cycles. If the stem weight gains measured in this experiment can be confirmed under field conditions, the length of these payback periods could be further reduced, since in this study the average cane price (*Cp*) was set to AU$ 500 t cane^{-1} but in 2022 and 2023 the average sugarcane price increased to above AU$ 600 t cane^{-1} and AU$ 900 t cane^{-1}, respectively (QSL 2023).

Globally, natural zeolites are highly abundant and are mined in countries such as U.S.A., China, Turkey New Zealand and Australia, which alone extracts approximately 10,000 to 20,000 t per annum. The price of natural zeolites (AU$ 240 t^{-1}) is roughly double that of other soil amendments that correct pH and nutrient availability, such as lime (AU$ 100 t^{-1}) or gypsum (AU$ 90 t^{-1}). However, the economic sustainability of natural zeolites is supported by the fact that, unlike lime or gypsum, zeolites do not break down over time. This means that their addition to agricultural soils could provide a permanent increase in their CEC and therefore their NH_4^+ retention capacity. Reduced N losses and increased N retention in the plant-soil system have been reported over multiple seasons after the application of zeolites (Ferretti et al. 2016), indicating that sugarcane farmers might have to apply zeolites only occasionally to counteract the dilution effect caused by tillage. However, future work will also have to consider the 'recharge' potential of zeolites and assess whether they retain and then release N consistently season after season, or become NH_4^+ or cation-saturated. If the latter case, it would depend on how long it takes NH_4^+-saturation to occur, and the quantities of NH_4^+ that can be exchanged in response to plant demand.

Study limitations

The aim of this study was to provide an indication of the *potential* environmental, agronomic and economic implications of using zeolites in sugarcane systems to guide future field experiments. The considerations made in this article are therefore not meant to be used as an exhaustive farm-gate analysis but should be evaluated taking into account the following limitations.

Firstly, this study was conducted under controlled conditions in a glasshouse to test the use of zeolites under the most extreme pedoclimatic conditions: sugarcane grown on a sandy-loam soil during the wet summer months of the Wet Tropics. The extent of the environmental (e.g. N loss reduction), agronomic (e.g. yield increase) and economic (e.g. extra income) benefits under less extreme conditions can therefore be expected to be less pronounced than those reported in this study.

Secondly, the zeolite application rate used in this study was defined to raise the CEC of the topsoil from to 1.6 cmol(+) kg soil^{-1} to 50 cmol(+) kg soil^{-1}. Under field conditions, the required zeolite rates might be lower since the starting CEC value of the topsoil may be higher than in this study, or it might prove sufficient to increase the CEC to 20-30 cmol(+) kg soil^{-1} to achieve satisfactory N loss reductions and yield improvements - possibly without reduced availability of other nutrients.

Lastly, since the glasshouse experiment lasted only eight months, in this study it was not possible to estimate the longevity of the environmental and agronomic benefits that could be expected after the plant crop. The CEC analysis of the soil layers indicated that both zeolite types did not move down the soil profile (Fig. 5), and therefore they would have been available to bind the excess N applied in the topsoil during the following ratoon crops. However, subsequent harvest, tillage and planting operations might scatter the zeolites on the soil surface or displace them down the profile, away from where the N fertiliser is applied. Moreover, once the zeolite sites are saturated with cations, their N adsorption capacity will be reduced and a new application might be needed.

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Implications for the sugarcane industry and future research needs

This study highlighted that zeolites have substantial potential to improve the sustainability and profitability of sugarcane cropping systems. In general terms, the sugarcane industry could incorporate the application of zeolites in their standard management practices in the following ways:

1. Acid-treated zeolites could be employed only in sugarcane systems characterised by high risk of elevated N losses, where water quality is a priority over yields (e.g. close to the Australian Great Barrier Reef). The adoption of this strategy would most likely require financial incentives, such as nutrient offsets or credit schemes to compensate for potential yield penalties.
2. Natural zeolites could be applied in any sugarcane system with high rainfall conditions (e.g. Wet Tropics, Humid Subtropics) and/or with coarse soil textures (e.g. sand content >40%) to reduce fertiliser N rates and N losses, while maintaining or even increasing cane yields.
3. To ensure the shortest payback period, zeolites could be applied in 20 cm wide strips around the fertiliser band, where the majority of mineral N will likely accumulate and therefore be prone to loss via leaching, runoff or denitrification.

However, this study was conducted under controlled environmental conditions, and future research should validate these results using multi-seasonal field studies conducted across different pedoclimatic regions. Specifically, future research should focus on addressing the following research questions under field conditions:

i. The longevity of the effects connected with the application of zeolites: how much will planting, tillage and harvest operations affect the spatial distribution and concentration of zeolites in the root zone? Will zeolites move throughout the profile or remain on the surface? Will it be necessary to apply zeolites at the beginning of every crop cycle (5 years)? Is applying zeolites in 20 cm wide strips around the fertiliser band ($Fa = 10\%$) going to be feasible under field conditions? Will the zeolite remain in such bands over multiple crop cycles?
ii. The ability of zeolites to sustainably reduce current fertiliser N rates in the long-term: how much N fertiliser rates could be reduced without resulting in yield penalties and mining native soil N? How much will N losses be reduced across different soil textures? How much of the N adsorbed by the zeolites will be made available in the subsequent seasons?
iii. The excessive efficiency of acid-treated zeolites in retaining macro- and micronutrients: once the negatively-charged sites on the zeolite surface are saturated with cations, will nutrient immobilisation still be an issue, and will fresh NH_4^+ still be adsorbed efficiently?
iv. The economies of scale: will the extra income generated by the zeolite application be confirmed over multiple cropping cycles under commercial settings?
v. The corollary environmental benefits: aside from reducing N losses via leaching, can the capacity of zeolite to adsorb and retain excess N be used to reduce N_2O emissions? Can such practices contribute to reduce the carbon footprint of the sugarcane industry by reducing the N fertiliser rates and associated Scope 1 and 3 greenhouse gas emissions?
vi. The statistical significance of the trends identified in this study: will large datasets collected in the field over multiple seasons validate the environmental and agronomic performances of natural and acid-treated zeolites?

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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